

Facilitating digital accessibility for students with disabilities into information services at Jordanian academic libraries

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Abstract

Purpose – This study examines the digital accessibility of information services for students with disabilities at the University of Jordan. Despite advancements in disability rights and technology, academic libraries often lag in providing fully accessible digital services.

Design/methodology/approach – The mixed-methods research involved quantitative surveys from 173 students with disabilities and a qualitative interview with the head of the information application department, in addition to a focus group interview with 10 students with disabilities. Quantitative analysis included means, standard deviations and four-way ANOVA to identify significant differences in perceptions. Qualitative insights highlighted specific issues and recurring themes, revealing significant gaps in digital accessibility.

Findings – The results revealed a weak level of digital accessibility of students with disabilities to information services in academic libraries in Jordan is very low. The findings indicate a need for improvements in library infrastructure, staff training and policy development. The study also offers insights into challenges in a developing country context and provides recommendations for enhancing library inclusivity and support structures, emphasizing the importance of aligning services with international accessibility standards.

Practical implications – The findings will primarily be beneficial for library managers to understand their library's deficiencies and responsibility towards the local community and the enhancement of digital inclusion. This understanding will aid in planning training programs and workshops for employees on supporting students with disabilities. Additionally, it serves as a valuable resource for collaboration between libraries and educators to organize educational sessions for academic librarians in Jordan and other developing nations.

Originality/value – It serves as a valuable resource for collaboration between libraries and educators to organize educational sessions for academic librarians in Jordan and other developing nations. This research adds value to existing literature by highlighting the context in one developing country.

Keywords Academic libraries, Digital inclusion, Information services, Digital accessibility, Students with disabilities, Integrative environments

Paper type Research paper



Introduction

Diversity and inclusion remain pressing topics in higher education and broader society, shaped by accreditation standards, laws and social change agents (Brannen *et al.*, 2017). Libraries, as service organizations, cater to a diverse clientele, including individuals with disabilities, offering nondiscriminatory access to resources. With the rise of the Internet as a

global information hub and the increased number of disabled students enrolling in higher education, libraries are prompted to reassess and ensure their resources and services are equally accessible to all users (Ekwelem, 2013). This includes maintaining a variety of accessible formats for library materials (Queiros *et al.*, 2015). The evolution of digital technologies and electronic resources has bolstered libraries' roles in ensuring all individuals can access their services, aligning with the United Nations principles on the inclusion of people with disabilities in all life aspects (Mladenov, 2013).

With the Americans with Disabilities Act (ADA) emphasizing access to library services, libraries have been key in bridging the digital divide by providing technology and digital literacy programs. Studies by Wiler and Lomax (2000) and Schmetzke (2001) evaluate the ADA's effectiveness, particularly for services aiding the blind and visually impaired. The UK's Special Educational Needs and Disability Act (SENDA) 2001 similarly focuses on accessibility compliance. Beyene (2018) highlights the need to thoroughly explore digital accessibility, considering access, skills, participation and usage to advance the goal of universal access.

Academic libraries are recognizing the importance of planning and evaluating their services, especially those for people with disabilities for a comprehensive inclusion of all different library users (Kirillova, 2017). Information services are essential not only for treating individuals with disabilities on par with others but also for enhancing their cultural, mental and scientific capacities, thus mitigating the impact of their disabilities and promoting full societal inclusion. Calvert (2008) adds that improving library services for students with disabilities requires clear policies and procedures, staff development and training, equal access to facilities and services and strategically placed specific technology support, often in an adaptive technical room within the library. Together, these measures help bridge the gap in access to information for individuals with disabilities. Libraries use many digital technologies, including web pages, where when designing the library website, it must be considered that access to the page by persons with disabilities must be facilitated by using a range of supporting devices and programs. It must, therefore, be designed in an effective way using supporting options and programs to provide persons with disabilities easy access to information (Billingham, 2014). Also, when the library selects databases and programs, it should take into account the ability of persons with disabilities to use these programs and services and provide assistive tools such as keyboards. Also, the availability of options that enable them to adapt to their nature, such as font size should be supported. Supporting techniques to show them their location on the screen, and tools to properly describe program functions and support commands with the appropriate image should be available for people with disability.

Research problem

Despite legislative frameworks in Jordan ensuring the rights of individuals with disabilities to access education and information, academic libraries, particularly at the University of Jordan, face significant challenges in providing adequate digital accessibility. Existing research indicates systemic deficiencies in supporting students with disabilities, particularly regarding digital access and resources. While some measures have been taken to improve accessibility, there remains a notable gap in fully integrating and implementing these services effectively.

The primary issue prompting this study is the substantial gap between the legal requirements for accessibility and the actual experiences of students with disabilities in academic libraries in Jordan. Although the Law on the Rights of Persons with Disabilities No. 20 of 2017 mandates accessible educational and informational resources, and the UN Convention on the Rights of People with Disabilities outlines the necessity for inclusive

facilities, there is still a lack of comprehensive implementation in digital library services (United Nations, 2016; The Law on the Rights of Persons with Disabilities No. 20 of year 2017, 2017). Furthermore, previous studies have highlighted various barriers faced by students with disabilities, including inadequate access to digital sources, insufficient support and training and a lack of specialized devices and programs (Beyene, 2018; Smadi, 2022). The University of Jordan, despite being a leading institution, has not fully met the accessibility needs of its disabled students, indicating a broader issue within Jordanian academic libraries (Hamad, 2023).

By investigating the digital accessibility of information services at the University of Jordan, this study aims to identify specific barriers and propose actionable solutions to enhance inclusivity. The findings will contribute to the literature by providing a detailed analysis of the challenges and recommending strategies for improving digital accessibility in academic libraries in a developing country context (Alabi and Mutula, 2020; Ayoung *et al.*, 2021; Phukubje and Ngoepe, 2017). In addition, the challenges faced by academic libraries in enhancing digital accessibility are multifaceted, involving issues such as the lack of diversity and inclusion in library collections (Gonzalez, 2024), and ensuring fair access and customization of educational resources (Uddin *et al.*, 2024). Additionally, e-learning facilities often suffer from accessibility issues (Ogunwuyi, 2024). Inadequate digital infrastructure hinders the effectiveness of academic libraries, particularly in regions like Africa (Adarkwah *et al.*, 2024). Adapting to new technologies, such as virtual reality, and ensuring sustainable access to digital content also present significant challenges (Tiloooy *et al.*, 2024). These challenges necessitate comprehensive strategies and innovative solutions to improve digital accessibility in academic libraries globally.

This study proposes to investigate the extent of digital accessibility in the services provided by the academic libraries at the University of Jordan, focusing particularly on the experiences and perspectives of students with disabilities and service providers. This research will address fundamental questions related to the accessibility and effectiveness of digital library services for this demographic, aiming to identify and mitigate existing barriers to their full participation in the academic community. This research aims to address key questions, including:

1. What are the experiences and perceptions of students with disabilities regarding digital accessibility in the information services provided by the University of Jordan's library?
2. Are there any differences in the perceptions of digital accessibility to information services among students with disabilities based on gender, academic year, cumulative GPA, type of college and nature of disability that are statistically significant at the 0.05 level?
3. What is the perception of the head of the information application department at The University of Jordan library of digital accessibility of information services for students with disabilities?

Literature review

Disability and digital access to information

The group of persons with disabilities is part of society that cannot be excluded or ignored. They are part of the groups with special needs (Foley and Ferri, 2012). The concept of persons with special needs from an educational perspective indicates that every person is deviated by his or her mental and physical characteristics, sensory abilities and social behavior from the average individual, and includes many categories, such as talented people, learning

difficulties, language disorders, persons with mental, visual, auditory, impaired mobility and emotional disabilities. Since the old decades, the issue of persons with disabilities has been and is still a core issue and a national priority. In Jordan, efforts to enable persons with disabilities to have full access to their rights through the enactment of laws, legislation and strategies preventing any form of marginalization of people with disabilities are continuous. The proportion of persons with disabilities in Jordan ranges from 3–5% of the total population (Al-A'dra, 2016). Al-Khatib (2017) indicates that one most important right of people with disabilities is their right to full and equal participation in all aspects of society. Because access to knowledge and information is a right for all, and because “information justice is one of the fundamental principles and requirements of public libraries as stated in the 1994 UNESCO statement,” it was important that information and communication technology be used in academic libraries as well (Zhao *et al.*, 2019).

People with disabilities are individuals who suffer long-term physical, mental, intellectual or sensory vulnerabilities that may impede individuals from overcoming various obstacles or participating completely and effectively in society on an equal footing with others (Al-Khateeb *et al.*, 2015). Strategies that aims at providing opportunities (training and services) to develop digital skills and understanding, which are designed to meet the challenges, faced by the digitally disadvantaged person is essential as means of accessing digital resources, training or other opportunities (Jaeger *et al.*, 2012). Facilitating digital accessibility is all of the motivational practices of using digital media in most life activities, such as viewing content through a new and interesting method, sometimes interactive, often through the use of smartphones, whether at home or in different places (Kaufman and Horton, 2014). This means that data and information on a particular electronic device can be read or processed by another device, using a particular format and application, and this service is essentially done; to facilitate access to information and cognitive data from different places, using computers or smartphones (Chen and Li, 2022).

The use of information technologies creates new channels for people with disabilities to access information and participate in social; and scientific activities that can be difficult in the real world, it removes obstacles they may face, and provides opportunities for learning, education and inclusion (Jaeger *et al.*, 2012). Equal access to and involvement in the online world is now a must for education, work, finance and civic engagement due to the omnipresent nature of the internet and its attendant services and technologies (Jaeger, 2011). The National Center for Support to People with Disabilities in the United States, emphasizes the importance of technology in empowering people with disabilities, as technology enhances information practice for most people and people with disabilities (Raja, 2016).

During the past decades, the pursuit of a comprehensive education system for people with disabilities has been one of the main demands. This means guarantees the right to and access to education and the right to full participation in society for individuals with disabilities (Soares and Faria, 2015). Disabilities come in a variety of forms. The International Classification of Functioning (ICF, 2001) defines disability as difficulties encountered in any or all three areas of functioning related to impairments, activity limitations and participation restrictions. People with disabilities are defined as individuals who suffer from a permanent state of physical or mental morbidity, which prevents them from fully and actively participating in society, and impedes their ability to adapt to things as healthy persons, thereby achieving equity between them and others (ICF, 2001). Individuals with disabilities may suffer from numerous disabilities such as hearing or visual disabilities, mental developmental delays, which may cause learning delays, behavioral disorders, psychological disabilities, language disorders and other cases. The term “persons with disabilities” explains the existence of a fundamental difference from the average or normal persons, as it is a term given to a person who is different from ordinary people in terms of mental, physical or sensory capabilities, or in terms of behavioral, linguistic or educational characteristics,

which require providing special education and support services; To meet the unique needs of the person (Al-Khateeb and Hadidi, 2009).

Digital accessibility in academic libraries

Access to higher education by people with disabilities has forced a qualitative shift in academic libraries, forcing them to advance to other levels, such as the generation and use of information in research activities conducted in institutions of higher education (Diniz *et al.*, 2019). The International Federation of Library Associations (IFLA) part of its mission is to “empower and inspire society by advancing access to information, knowledge, and culture for everyone, to encourage growth, learning, creativity, and innovation.”. Its four core values reflect this dedication and emphasize that high-quality library services should be accessible to everyone, including those with disabilities. As the world’s largest organization representing all sorts of libraries, IFLA has a special focus on libraries for persons with special needs, as evidenced by the Marrakesh Treaty (2016), the CRPD and the Sustainable Development Goals (IFLA, 2023). The Marrakesh Treaty aims to improve the availability and cross-border exchanges of certain works for visually impaired or otherwise print-disabled individuals, ensuring that they have access to the same information and knowledge as sighted individuals (WIPO, 2016). This goes in harmony with the UN initiative on Monitoring and Evaluation for Disability-Inclusive Development (Bolt and Wyber, 2017).

According to the American Library Association Council (ALA, 2015), libraries play an active part in the lives of people with disabilities by enabling their full participation in society. Libraries should use universal design principles to ensure that their policy, resources and services meet the needs of all individuals. Incorporating universal design principles simply means that care is given when developing any service, program or instruction to ensure that it is accessible to the broadest possible audience without the need for additional changes or considerations (Weber and Yarnetsky, 2024). Furthermore, the United Nations Convention on the Rights of Persons with Disabilities (United Nations, 2016) in Article 24 affirmed that: “persons with disabilities are able to access general tertiary education, vocational training, adult education, and lifelong learning without discrimination and on an equal basis with others. To this end, States Parties shall ensure that reasonable accommodation is provided to persons with disabilities”.

Alabi and Mutula (2020) study highlights significant barriers that visually impaired students face, such as accessing printed materials and navigating physical spaces, and discusses how assistive technologies like smart canes, wearable ultrasonic devices and smart glasses that can aid in mobility, information access and interaction with digital content. Ayoun *et al.* (2021) found that most libraries in Ghana are not fully accessible beyond the main entrance and lack diverse media for people with visual and hearing impairments. Additionally, the study highlights that library staff generally lack the knowledge and training to assist persons with disabilities effectively. The study recommended empowering relevant institutions to enforce the Disability Act more vigorously and increasing investment in library services tailored for PWDs. Phukubje and Ngoepe (2017) assessed the accessibility and convenience of library services provided to students with disabilities at the University of Limpopo, South Africa. Despite having a purpose-built library service unit that complies with international standards, findings indicate that students with disabilities are not fully satisfied with the services received. Key issues include inadequate transcription of materials into accessible formats and insufficient staffing, highlighting the need for better resource allocation and staff training to enhance library service accessibility for these students.

Roberson *et al.* (2022) emphasizes the importance of inclusive library services that integrate both universal design and specific accommodations, viewing inclusion not as a

static achievement but as an ongoing process of adaptation and engagement with students' needs. The study also suggests practical recommendations for libraries to become more inclusive, such as increasing librarian involvement in classroom settings and using Universal Design principles to enhance accessibility and usability for all students. [Onsinyo \(2018\)](#) focuses on the significant barriers that persons with disabilities face in accessing library and information services in university libraries. Despite the presence of such services, the study identifies that these are largely inaccessible to PWDs due to structural inadequacies, unskilled staff and a lack of specialized facilities like lifts, spacious ramps, Braille materials and hearing aids. The research advocates for a more inclusive approach in the design and operation of library services that cater effectively to the needs of PWDs, emphasizing the importance of training for library staff and the integration of appropriate assistive technologies. [Smadi \(2022\)](#) examined the availability and usage of various services and facilities within the university's central library from the perspective of students with disabilities at Imam Muhammad bin Saud Islamic University in Saudi Arabia. Utilizing a descriptive survey, it finds that while some services like book borrowing and return, as well as audiobooks and Braille services, are highly utilized and rated, other areas, especially related to physical accessibility and technological aids, are lacking and rated poorly by the students. The study underscores the need for enhanced services that cater specifically to the needs of students with disabilities, aiming to provide them with more equitable access to educational resources.

Moreover, the ability to read all texts in programs and databases using supporting techniques, and the existence of graphics-equivalent text, the ability to modify color settings and views without affecting the efficiency of programs and databases, and the possibility of entering information using supporting techniques into databases and programs, all are needed as supporting devices and equipment to support digital access to information ([Mates and Reed, 2011](#)). [Irvall and Nielsen \(2005\)](#) and [Bellingham \(2014\)](#) added that libraries must design their pages logically and easily to facilitate their use by disabled individuals. Websites need to incorporate font and text characteristics with modification options and provide texts with appropriate comparative measurements for persons with disabilities. Studies have pointed out that the combination of assistive technologies and specialized database systems allows dramatically improving the accessibility issue of library services provided to students with disabilities ([Bulao et al., 2022](#)). The relevance of adaptive tools and individualized support is confirmed thus creating a common idea to select participative strategies.

[Pionke \(2020\)](#) revealed that library staff at the University of Illinois Library in Urbana-Champaign are reluctant to work with disabled individuals and highlighting the need for significant modifications in how libraries serve this students category. Furthermore, the study emphasized the necessity for developing more inclusive spaces and programs. [Ilako et al. \(2020\)](#) found that despite recognizing the library's service contributions, many students noted a deficiency in modern academic resources in accessible formats like Braille or digital text. [Bhardwaj \(2018\)](#) has pointed out that academic libraries in Delhi, India, provide only basic facilities for students with visual impairments, with inadequate information and communication technology infrastructure to support their research and study needs. In addition, [Hamad \(2023\)](#) revealed a significantly low level of digital accessibility of information services at The University of Jordan for students with disabilities. Previous research has largely focused on the perspectives of library staff, with limited studies incorporating students' views or specifically targeting the needs of people with disabilities. This scarcity of research extends to academic libraries in Jordan, where studies on digital accessibility for individuals with disabilities are notably absent. This current study aims to fill these gaps by differing in its objectives, demographic focus, geographical setting and timing.

Digital accessibility for information services is a critical area of research, particularly for students with disabilities. Various studies have highlighted the challenges and potential solutions in this domain. One significant challenge is the lack of diversity and inclusion in library collections and services. [Gonzalez \(2024\)](#) highlights that many academic libraries conduct diversity audits to identify gaps and improve digital accessibility for diverse user groups. These audits are crucial in ensuring that collections reflect the diverse backgrounds and needs of the library's users, promoting an inclusive environment. [Banerjee et al. \(2024\)](#) discuss the development of a machine-learning-based accessibility system designed to enhance the usability of digital platforms for users with disabilities. Their research demonstrates the potential of advanced technologies in breaking down barriers to digital accessibility. Also, ensuring fair access to educational resources is another critical issue. [Balamani et al. \(2024\)](#) focus on empowering rural students with physical disabilities through smart learning environments. They emphasize the importance of flexible and accessible educational tools in bridging the gap between urban and rural education systems. [Uddin et al. \(2024\)](#) discuss the development and assessment of Open Educational Resources (OER) materials, which are designed to be accessible and adaptable. These materials help in meeting the diverse needs of students, ensuring that all students have access to high-quality educational content regardless of their socio-economic status. [Prince \(2024\)](#) critically examines the content of library school syllabi, revealing a significant lack of focus on disability and accessibility issues. This study calls for a comprehensive overhaul of educational curricula to include more content on digital accessibility and disability accommodation. [Zaid \(2024\)](#) underscores the critical role of libraries in providing equitable access to learning resources for students with disabilities. The study advocates for enhanced library services that cater to the diverse needs of all students. [Hsu and van der Hof \(2023\)](#) suggest that thoughtful design of digital products and services can significantly enhance accessibility and user experience for individuals with disabilities.

High costs associated with acquiring digital materials, such as audiobooks, pose a significant barrier to accessibility. [Silva \(2024\)](#) addresses this challenge by discussing initiatives for textual representation and cost-effective access to information resources. By focusing on affordable solutions, libraries can expand their digital collections without placing undue financial strain on their budgets. [Ogunwuyi \(2024\)](#) examines the accessibility issues in e-learning facilities and digital libraries, particularly in Nigerian colleges. The study emphasizes the need for improving the availability and effective use of digital libraries to support e-learning and teaching processes. Enhancing digital infrastructure and training educators on effective utilization of these resources are critical steps in overcoming these challenges. Inadequate digital infrastructure hinders the effectiveness of academic libraries, particularly in regions like Africa ([Adarkwah et al., 2024](#)). The integration of new technologies, such as virtual reality (VR), presents both opportunities and challenges for digital accessibility. [Tiloooby et al. \(2024\)](#) examine how VR can revolutionize academic library services and improve digital accessibility. Librarians need to adapt to these new technologies to provide sustainable access to digital content and enhance the user experience.

The integration of advanced technologies, inclusive design principles and comprehensive educational reforms are essential to create accessible and equitable information services. Continued research and innovation are necessary to address the evolving needs of students with disabilities and ensure their full participation in academic and social life.

Methodology

This research investigates students with disabilities experience of digital access to information services in academic library services at the University of Jordan. This research is

designed based on a case study design choosing The University of Jordan as the case study as it is the leading university in the country. The study adopted a mixed research approach. Exploratory sequential design was employed starting with quantitative methods to investigate students with disability experience of digital accessibility of information services in the library. Then qualitative method was used to explore and explain the quantitative results.

Study population

The study community consisted of all students with disabilities ($N = 284$) who were enrolled at the University of Jordan during the academic year (2022/2023). This number was obtained from the Department of Admission and Registration at the University of Jordan. There were many types of disabilities. Students were reached through the Student Guidance Department/Division of Special Needs in the Deanship of Student Affairs. A questionnaire was distributed to all students. Of the 284 students, 173 male and female students with disabilities responded, and their responses were suitable for analysis with a response rate of (61%). Table 1 demonstrates the distribution of the students based on several factors that might affect respondent perceptions of digital accessibility in information services at the University of Jordan Library. The varying factors are Gender, nature of disability, college, academic degree, academic year, GPA. Furthermore, to answer the third research question, the head of the information applications department was interviewed to discuss and elaborate more on the quantitative results. The head of the department was asked to discuss and comment on the results from the quantitative phase to provide more in-depth explanations and insights related to the findings. Additionally, a focus group consisting of 10 students with disabilities was conducted to gather detailed feedback on their experiences and challenges with digital accessibility in the library.

Factor/Variable	Variable type	No.	%
Sex	Male	71	41.0
	Female	102	59.0
	Total	173	100.0
Disability type	Motor	53	30.6
	Hearing	86	49.7
	Visual	34	19.7
	Total	173	100.0
College	Humanities Colleges	147	85.0
	Science Colleges	26	15.0
	Total	173	100.0
Educational level	Bachelors	143	82.7
	Postgraduate	30	17.3
	Total	173	100.0
Academic year	First	38	22.0
	Second	57	32.9
	Third	46	26.6
	≥Fourth	32	18.5
	Total	173	100.0
GPA	3.65–4	41	23.7
	3–3.65	98	56.6
	≤2.99	34	19.7
	Total	173	100.0

Table 1.
Study population

Research questions

To answer the study questions, a questionnaire was developed; aiming to identify the experiences and perceptions of students with disabilities regarding digital accessibility in the information services provided by the University of Jordan's library. The first part of the questionnaire collected personal demographic information to explore the differences in the perceptions of digital accessibility among students with disabilities based on gender, academic year, cumulative GPA, type of college and nature of disability that are statistically significant at a significance level of 0.05. The second part of the questionnaire consisted of four main domains in an attempt to answer the main research questions, which are:

- (1) First domain: Access to digital sources of information.
- (2) Second domain: Devices and programs to support persons with disabilities.
- (3) Third domain: The Support options provided for persons with disabilities.
- (4) Fourth domain: The difficulties faced by students with disabilities in information services at the University of Jordan library.

A Likert scale was applied to weight the degree of agreement with the questionnaire items. Table 2 demonstrates the range of weights used to determine the degree of agreement for each item using the mean value.

Questionnaire validity. The questionnaire was reviewed by a group of referees from the faculty of Educational Sciences and library, including faculty members, specialists and expertise in library science, information and special education; to check for clarity and grammatical errors and to identify the suitability of the questionnaire items to provide answers to the two main questions. Their views and suggestions were taken into account, by making some of the recommended adjustments, as they saw fit, whether by deletion, consolidation or addition, in order to improve the study tool, as the approval rate (80%) and more was adopted on the paragraph as evidence of its validity. Moreover, a Cronbach Alpha test was conducted to ascertain the reliability of the questionnaire. The Cronbach Alpha value was high, scoring 0.87 as indicated in Table 3.

Table 3 shows that Cronbach Alpha's coefficients for all domains ranged between (0.81–0.95), all of which are greater than (0.70), demonstrating the validity of the tool to achieve the objectives of the study.

Table 2.
The weights used to
determine the degree of
agreement for
each item

Degree	Mean
High	3.68–5
Moderate	2.34–3.67
Weak	1–2.33

Table 3.
Cronbach Alpha test

Factor	No. of items	Cronbach Alpha
• Access to digital sources of information	10	0.87
• Devices and programs to support persons with disabilities	16	0.95
• The support provided for persons with disabilities	7	0.81
• The difficulties faced by students with disabilities in information services at the University of Jordan library	13	0.86
<i>Total Cronbach Alpha</i>	46	0.87

Statistical analysis

Using SPSS, mean and standard deviation were calculated to assess the degree of respondent's perception of digital accessibility in information services at academic library using the University of Jordan as a case study. FOUR-WAY ANOVA analysis was conducted to investigate the existence of any statistical differences, at a significance level ($\alpha = 0.05$) using *f*-tests between responses based on different variables. The focus group data was analyzed using thematic analysis, a qualitative research method that involves identifying, analyzing and reporting patterns (themes) within the data. Following this systematic approach, the focus group data was thoroughly analyzed, providing valuable insights into the digital accessibility challenges faced by students with disabilities at the University of Jordan.

Results and discussion

Digital accessibility of information services

To answer this question mean and standard deviation (SD) were calculated for all domains under the main research question to understand the student's perception of digital accessibility in information services at the University of Jordan library as an example of academic libraries in Jordan. Table 4 presents ratings for various aspects related to the accessibility and support services for students with disabilities in the University of Jordan library.

The results indicated in Table 4 that the students rating of digital accessibility of information services at The University of Jordan library was low with a mean that leans towards the lower end of the scale ($M = 1.74$; $S.D = 0.45$). The relatively lower standard deviation for the total degree indicates a uniformity in the perception of the participants regarding the general state of digital accessibility in the library. One of the domains related to the students assessment of the difficulties faced by students with disabilities in information services at the University of Jordan library was rated moderate, ($M = 3.58$; $S.D = 0.62$) while all other domains ratings were low. The relatively higher mean suggests that while there are challenges, they might not be as severe as completely hindering access. The moderate standard deviation implies that responses were somewhat varied but not extremely so.

Regarding access to digital sources of information, the rating was extremely low ($M = 1.97$; $S.D = 0.57$), indicating that students with disabilities perceive access to digital sources as inadequate, reflecting a significant area of concern. The low standard deviation suggests that responses were relatively consistent among participants, reinforcing the concern over access to digital information. As for students experience with the support provided for persons with disabilities is rated low, similar to accessibility issues reported earlier. This indicates a perceived lack of sufficient support for students with disabilities. The standard deviation is tight, suggesting a general agreement among respondents about the inadequacy of support. Furthermore, students rating for the availability of special

Table 4.
Means and standard
deviations (SD) of
responses on students
perception of digital
accessibility in
information services at
the university of
Jordan library

Item No.	Factor	Mean	S.D	Rating
4	The difficulties faced by students with disabilities in information services at the University of Jordan library	3.58	0.62	Moderate
1	Access to digital sources of information	1.97	0.57	low
3	The support provided for persons with disabilities	1.95	0.52	low
2	Special devices and programs to support persons with disabilities	1.54	0.51	low
	Total Degree	1.74	0.45	Low

devices and programs to support persons with disabilities was the lowest among all other domains ($M = 1.54$; $S.D. = 0.51$), suggesting that the respondents perceive a significant deficiency in special devices and programs designed for persons with disabilities.

The findings collectively indicate a systemic deficiency in supporting students with disabilities at the University of Jordan Library, especially concerning digital access and resources. Despite moderate difficulty levels suggesting problems are addressable, the low overall evaluations, particularly in digital access, support services and device availability, underline a critical need for enhanced digital accessibility and academic support. The situation reflects broader trends identified in regional studies, showing a lack of specialized information sources and inadequate facilities for visually impaired students in higher educational institutions across the Middle East (Ilako *et al.*, 2020; Bhardwaj, 2018). Fernández-Batanero *et al.* (2022) review the systematic role of assistive technologies in promoting inclusion for students living with disabilities.

Qualitative results attribution. Investigating the library strategic plan confirmed that the library plan in general focuses on superior information service in terms of collection quality, availability and accessibility to foster creativity and a sustainable educational environment for its users. The strategy does not contain an explicit or clear policy for information services tailored for students/people with disabilities that confirm student report experience (The University of Jordan Library, 2023).

To further understand these results, they were discussed with the head of the information application department. The discussion lasted for 45 min. The director acknowledged the lack of sufficient support provided to the students with disabilities at their library. He acknowledges specific deficits in the availability of adaptive technology, accessible digital content and training for staff and students, in this regard, the director said, “*We’ve identified several key areas where support is insufficient, including adaptive technology, accessible digital content, and specialized training for both staff and students.*” Recognizing specific areas is the first step toward addressing the gaps, showing a focused and informed strategy, which in turn suggests a targeted approach to problem-solving. Apparently, the University of Jordan library relies on the efforts of volunteers from ordinary university students to assist and support students with disabilities to benefit from the services and resources provided by the library. Burrows (2019) supported this action to ease the workload of the library staff and provide one-on-one support for students with disabilities. Furthermore, Williams (2017) affirmed the need for collaborative work between faculties, schools and departments to develop accessible learning resources. This will familiarize library staff with thinking about teaching practice leading them to be creative in tailoring more accessible assistive technologies to a wider range of students with disabilities. It is always crucial to raise awareness of accessibility and provide training related to inclusive teaching and learning.

The director also signifies the need for a multi-faceted approach to confront the insufficient support they provide for students with disabilities. By investing in technology, improving physical accessibility and enhancing staff training, the response demonstrates a commitment to holistic improvement (MacLachlan *et al.*, 2018; McGowan *et al.*, 2018). This proactive approach might reflect a willingness to invest in resources as required as he mentioned “*We are committed to addressing these gaps to ensure equitable access to all our resources.*” The director elaborated more “*Certainly, we have recently introduced text-to-speech software in the computers in the specialized room for disabled students and have begun offering assistive technology training sessions.*” Affirming the library initiatives to provide a special room for people with disabilities. The recent initiatives by the library such as text-to-speech software and assistive technology training sessions demonstrate the proactive measures made to alleviate concerns. This means that the problem has been identified and is being actively addressed reflecting the library’s orientation toward an inclusive learning

environment. The statement of ongoing audits and compatibility with screen readers demonstrates a consistent dedication to diversity. However, is better to provide assistive software on all rooms and computers to ensure that students with disabilities have continuous access to all services (Williams, 2017).

Discussing issues such as funding and maintaining digital accessibility standards demonstrates transparency and knowledge of the ongoing nature of these initiatives. Recognizing these difficulties indicates an awareness of the complicated and ever-changing context of disability assistance services. The director proposed an all-inclusive approach to eliciting student feedback to assist and guide future enhancements to information services. Understanding the needs of impacted students directly through surveys, focus groups and advisory committees comprised of students with disabilities demonstrates the institution's commitment to participatory governance and continual development based on user experience. In the same vein, Seale (2009) and May and Bridger (2010) affirmed the importance of students inclusion and participation and needs assessments. Articulating a long-term vision centered around creating an inclusive environment and anticipating future needs illustrates strategic thinking. It is not just about resolving present challenges, but also about anticipating future changes. This forward-thinking mindset is essential for long-term and adaptive support services (Carabajal *et al.*, 2017).

Focus group findings

The findings from a focus group with students with disabilities regarding digital accessibility of information services in the library revealed several aspects that students think are important to enhance their inclusion in information services. The participants identified several accessibility issues within the library's digital services, highlighting challenges with screen readers, inadequate keyboard navigation and a general lack of compatibility with various assistive technologies. These issues reflect a need for continuous evaluation of digital platforms against current accessibility standards. In addition, a significant gap was noted in the availability of resources in accessible formats. Participants expressed a need for more audiobooks, large print materials, Braille and digitally navigable texts, indicating a discrepancy between available resources and the diverse needs of students with disabilities. There was a call for increased personalization of library platforms, including adjustable text sizes, contrast settings and customizable navigation options that retain user settings, enhancing the user experience for individuals with varying needs. Studies by Alabi and Mutula (2020), Ayoung *et al.* (2021) and Onsinyo (2018) consistently highlight significant barriers that prevent PWDs from fully accessing library services. These include difficulties in accessing physical spaces and digital content, particularly for those with visual and hearing impairments. Phukubje and Ngoepe (2017) and Smadi (2022) confirmed common issues include inadequate transcription of materials into accessible formats and insufficient utilization of available technologies.

Furthermore, the findings suggest an essential requirement for enhanced training programs for library staff centered on disability awareness and specific strategies to assist students with disabilities. Additionally, the establishment of training sessions for these students to effectively utilize digital resources was recommended, which could empower users and ensure equitable access. In addition, while the primary focus was on digital inclusion, comments regarding the physical accessibility of library facilities were also prevalent, underscoring how physical barriers can indirectly affect the utilization of digital resources. The participants advocated for more effective feedback mechanisms that allow for continuous input on library services, ensuring that modifications and improvements are responsive to user needs. There was a consensus that policies on inclusivity need clearer communication and greater transparency. This would ensure that all students are fully

informed of the accommodations and services available to them. [Ayoung et al. \(2021\)](#) and [Onsinyo \(2018\)](#) identify a general lack of knowledge and training among library staff on how to effectively assist PWDs. This deficiency underscores the need for enhanced staff training and resource allocation to improve service delivery. Furthermore, [Roberson et al. \(2022\)](#) suggest increasing librarian involvement in classroom settings and implementing Universal Design principles to make library services more inclusive and accessible to all students.

The focus group responses highlight an acknowledgment of the challenges faced by the University of Jordan's library, particularly regarding digital and physical accessibility for students with disabilities. The organization is actively working to address these issues, demonstrating a transparent approach that fosters trust and paves the way for effective problem-solving. Despite budget constraints, lack of strategic prioritization and insufficient staff training, which have been cited as major impediments, there is a concerted effort to improve service equality and accessibility. The discussion reveals an awareness of these challenges and a commitment to continuous improvement, emphasizing both immediate actions and long-term strategic planning to align the library's services with international standards of equality and inclusivity. By leveraging insights from the focus groups and collaborating closely with the head of information services, the library is poised to implement strategic changes that enhance service provision and foster an inclusive academic environment, thus supporting the academic success and social integration of all students, regardless of physical abilities. This proactive approach not only addresses current deficiencies but also sets a foundation for sustainable development and compliance with best practices in accessibility. Studies, including those of [Perez-Escolar and Canet \(2023\)](#), and [Hoyos Muñoz and Cardona Valencia \(2023\)](#), offer a broader insight into the challenges that digital inclusion faces including different policies and strategic planning to enhance the availability of digital resources to students with disabilities and their efficiency. [Ayoung et al. \(2021\)](#) recommend empowering relevant institutions to enforce the Disability Act more vigorously and increasing investment in tailored library services for PWDs.

Demographics effect on respondents experience

It is crucial to investigate the influential effect of external factors on students responses. Mean and standard deviation (SD) of responses based on the five variables as well as multi-way ANOVA were calculated. See [Tables 5 and 6](#).

Factor/Variable	Variable type	No.	Mean	SD
Sex	Male	86	1.70	0.380
	Female	130	1.76	0.490
Academic year	First	50	1.55	0.270
	Second	69	1.88	0.590
	Third	58	1.66	0.240
	≥Fourth	39	1.80	0.460
GPA	3.65–4	56	1.66	0.350
	3–3.64	118	1.75	0.460
	≤2.99	42	1.79	0.510
College nature	Humanitarian	170	1.70	0.430
	scientific	46	1.92	0.500
Disability nature	mobility	79	1.77	0.350
	Visual	106	1.68	0.460
	Hearing	31	1.83	0.540

Table 5.
Means and standard
deviations (SD) of
responses based on
different factors/
variables

The results revealed that the mean values of responses varied between respondents with different factors; gender, academic year, GPA, college nature and nature of disabilities. To support the result, multi-way ANOVA analysis at a significance level ($\alpha = 0.05$) between the means of responses of the students with disability based on different variables was conducted. The analysis suggested that there were no statistically significant differences at the significance level ($\alpha = 0.05$) between the means of responses based on sex, GPA and nature of disabilities variables. This could reflect that all students with disabilities have the same issues and concerns related to digital accessibility information services at the University of Jordan Library (see [Table 6](#)). This might reflect that the library's planning strategy for the delivery of digital services is a generic one without focusing on a specific category of users. It also indicates that the services are provided to everyone without prejudice to a certain category, level or nature, and they all receive the same services and are subject to the same laws and strategies.

However, the F-analysis results indicated the existence of statistically significant differences at the level of significance ($\alpha = 0.05$) between student assessment of their digital accessibility in information services at the University of Jordan library from their perspective, according to the variables collage nature (scientific, humanitarian) at significance level (0.047) and in favor of the scientific colleges, which had an arithmetic average of (1.92) compared to the humanitarian faculties (1.70). Also, the result indicated the existence of statistically significant differences at the level of significance ($\alpha = 0.05$) between student assessment of their digital accessibility in information services at the University of Jordan library from their perspective, according to the variables academic year at the significance level (0.009). Scheffe test showed that there are statistically significant differences in the estimates of students with disabilities of digital accessibility in information services at the University of Jordan library from their perspective, between first and second-year students, with a value of (−0.3293) and in favor of the second-year students, this indicates that the sample of second-year students believes that the digital accessibility of information services in the University of Jordan library is better than others, possibly because second-year students have become more aware of the reality of the services provided to students by the University of Jordan library. See [Table 7](#).

Research implication

The findings from this study on facilitating digital accessibility for students with disabilities in Jordanian academic libraries have significant implications for policy, practice and future research. First, the research underlines the critical need for libraries to reassess their digital and physical infrastructure to better accommodate students with disabilities. This involves not only upgrading technology and ensuring that digital resources are accessible but also training library staff to better support these students. These findings provide a clear directive for

Table 6.
Test analysis of
responses based on
different factors/
variables

Variance source	Total squares	Degree of freedom	Average squares	F value	Statistical significance
Sex	0.079	1	0.079	0.424	0.516
Disability nature	0.102	2	0.051	0.275	0.760
College nature	0.745	1	0.745	4.013	0.047
Academic year	2.211	3	0.737	3.972	0.009
GPA	0.011	2	0.006	0.031	0.970
Error	30.248	163	0.186		
Total	555.333	173			

library managers, to prioritize accessibility in both policy formulation and staff training. By understanding the specific barriers faced by students with disabilities, managers can implement more effective strategies that directly address these issues. Additionally, this study highlights the necessity for continuous improvement and adaptation of library services to keep pace with technological advancements and evolving accessibility standards. This study could serve as a benchmark for other academic institutions in similar regions or contexts, providing a comparative framework for assessing and enhancing digital accessibility.

Furthermore, the study underscores the importance of involving students with disabilities in the planning and implementation phases of library services. Their firsthand experiences are invaluable in identifying gaps in services and suggesting practical improvements. This participatory approach not only ensures that the services meet the actual needs of these students but also promotes inclusivity and a sense of community. In addition, for policymakers and educational leaders, the findings stress the importance of integrating accessibility into the broader educational framework. Ensuring that students with disabilities have equal access to information is not just a matter of compliance with legal standards but is also crucial for fostering an inclusive academic environment that values diversity and promotes equal opportunities for all students. This research, therefore, calls for a collaborative effort among library professionals, educational institutions, policymakers and the disabled student community to develop and implement comprehensive strategies that enhance digital accessibility and inclusivity in academic libraries.

Recommendations. Academic libraries play an integral role, socially and educationally, in digitally including students with disabilities in academic life however, the results of this research indicated that academic libraries did not fully realize their role and digitally include this category of students in their information services. The overall assessment of digital accessibility of students with disability into information services at academic libraries based on students perceptions was low. This suggests that academic libraries might still be at the beginning stages of realizing their role toward students with disabilities. Academic libraries in Jordan are focal in providing support to students with disabilities and facilitating their access to information. Accordingly, it is essential for these libraries to re-evaluate their inclusion policies to embrace more practices to embrace students with disabilities, as follows:

1. Preparation of comprehensive plans and programs with clear objectives by university administration; for comprehensive digital integration of students with disabilities.
2. Cooperating with the Deanship of Student Affairs at the university to organize regular meetings with students with disabilities to exchange views on problems and constraints about their integration in library information services. And also to discuss techniques for their development, and to provide them with the opportunity to select services, sources and equipment that meet their needs.
3. Establishing a specialized library unit or service to monitor the needs of students with disabilities and to provide support and guidance to them.

Table 7.
Scheffe test for means
between the means of
the respondents
estimates of the status
of digital accessibility
in information services
at the University of
Jordan Library from
their perspective
according to the
academic year

Academic year	Arithmetic averages	First	Second	Third	Fourth and more
First	1.5491	—	0.3293*	0.1151-	0.2557-
Second	1.8784	—	—	0.2142	0.0736
Third	1.6643	—	—	—	0.1406-
Fourth and more	1.8049	—	—	—	—

Note(s): *indicates at the level (0.05)

4. Adaptively design digital platforms, taking into consideration design issues and options that are suitable for students with disabilities.
5. Make more studies on needs assessments of students with disabilities to enhance the support strategies and digital accessibility practices that are employed by the library.
6. Working with publishers to create materials in a format, which is accessible for students with disabilities.

This spread the ground for academic libraries to be innovative in their practices and take action to embrace digital accessibility more. Although public libraries hold social responsibilities towards the community at the national level, also academic libraries share this responsibility with them. Based on the assessment of digital accessibility identified by respondents, libraries should re-evaluate their inclusion policies and advise more adjustment and adaptation of information services to support students with disabilities. They can also host local forums to discuss the challenges of digital accessibility from students perspectives and practical experiences. Furthermore, libraries need to create and design programs that can effectively contribute to enhancing students with disabilities' experience.

Conclusion

This research aimed to identify the status of digital accessibility of students with disabilities to information services at academic libraries in Jordan. Libraries are considered key institutions that guarantee access to information to all users with different categories and needs. The results of this research revealed that the digital accessibility of students with disabilities to information services in academic libraries in Jordan is very low. The University of Jordan did not provide the required support to this category of students. This calls for the need for academic libraries to revise their plans and policies to be more inclusive and to include students with disabilities in their information services planning. The study will be useful first for library managers to realize their library's role towards the local community and therefore the advancement of digital inclusion. This will help them to plan for their employee's training program and workshops on providing support to students with disabilities. Second, it will be useful for trainers to arrange training for academic librarians in Jordan and other developing countries.

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